

Project 3: Method Demo

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1. Introduction

In understanding a model, one needs to understand the limitations of the modeling method. Using a synthetic dataset that is intentionally designed to stress-test the method and a real-world dataset with similar structural challenge of little observations with multiple features, the logistic regression model demonstrates problems from overfitting variables and poor generalization, which is more apparent in synthetic datasets than in real-world datasets, and high computational intensity with more features, which is more apparent in the real-world dataset with a higher number of features than the synthetic dataset.

2. Project Dataset

The synthetic dataset is created with a small number of observations (100) and a large number of features (1000). A small subset of the features are truly predictive of the target variable, which is binary, and the majority of features are random noise, designed to confuse the model and induce overfitting.

The real-world dataset was found with the most features in the UC Irvine Database (Ma). The dataset has over 3.3 million features with 2.4 million instances. Using select days (day 0, day 1, day 100, and day 120) of the dataset, the instances or corresponding observations in the compiled dataset decreases to less than 20,000 per day. The features are anonymized to correspond with a feature at a certain index with certain values for the URL and the instances are binary classified into either a malicious URL (+1) or a benign URL (-1).

3. Methodology

Given a synthetic dataset with 1000 features with the majority being noise and only a few being truly predictive of the target variable, 100 observations, and a binary classification target, the modeling method, logistic regression, can be trained and tested on this split 0.3/0.7 dataset.

The Logistic Regression uses the logistic function:

$$p_{synthetic}(X) = \frac{e^{\beta_{0_{synthetic}} + \beta_{1_{synthetic}}x}}{1 + e^{\beta_{0_{synthetic}} + \beta_{1_{synthetic}}x}} \quad (1)$$

where $p_{synthetic}(X)$ is the logistical function, which creates an output between 0 and 1 that can be rounded down and up for classifying the target as 0 or 1, X is the input of 100 observations, each described by 1000 features, $\beta_{0_{synthetic}}$ and $\beta_{1_{synthetic}}$ are the unknown coefficients that the model finds using the equation of a line that best fits the synthetic data points.

Similarly, given the real-world dataset with 3.3 million anonymized features and 16,000 observations of URLs on Day 0, logistic regression can be trained and tested on the real world's split 0.3/0.7 dataset.

In the Logistic Regression function for the real-world dataset, it follows:

$$p_{real}(X) = \frac{e^{\beta_{0_{real}} + \beta_{1_{real}}x}}{1 + e^{\beta_{0_{real}} + \beta_{1_{real}}x}} \quad (2)$$

where $p_{real}(X)$ is the logistical function, which creates an output between 0 and 1 that can be rounded down and up for classifying the target as a benign URL (0) or a malicious URL (1), X is the input of 16000 observations, each described with 3.2 million features, $\beta_{0_{real}}$ and $\beta_{1_{real}}$ are the unknown coefficients that the model finds using the equation of a line that best fits the synthetic data points.

The real-world dataset is tested four times with Day 0, Day 1, Day 100, and Day 120 to see accuracy. It is tested with both a baseline model without any regularization and a model with regularization.

Using the training accuracy and the testing accuracies, including the average of the four Days, the percent error can be found to see if the training accuracy is significantly higher than the test accuracy, assuming anything at or below 5 percent is accurate. To find the percent error, we use the equation:

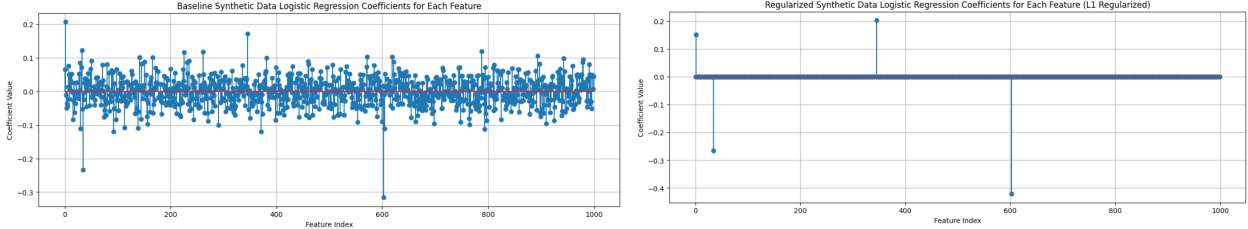
$$\%Error = \frac{actual - accepted}{accepted} \quad (3)$$

4. Results

In Table 1, the synthetic dataset show that the logistic model is trained to overfit with a 100% accuracy using all 100 features. Testing the model on the test set, it has 66.67% accuracy.

	B Accuracy	R Accuracy	Time / B % Error	Time / R % Error
Train	100%	88.57%	0.2464s	0.0076s
Test	66.67%	76.67%	33.33%	13.43%

Table 1: Baseline (B) and Regularized (R) Accuracy and % Error of Model and time required to train model using dataset using Synthetic Dataset



(a) Baseline: 1000 Features, 0.2133s to Graph

(b) Regularized: 4 Features, 0.2004s to Graph

Figure 1: Synthetic Data Coefficient Graphs

Considering the percent error, which is a 33.33% error, the baseline model can be considered significantly inaccurate (over 5% error). Comparing the regularized model’s accuracy on the training data, which is 88.57%, with the test date, which is 76.67%, the percent error decreased to 13.43%, but still significantly inaccurate since it’s over a 5% error.

According to Figure 1, the baseline model uses 1000 features, which allows the model to obtain 100% accuracy on the training set through overfitting, which when tested with the training dataset show a significant decline in accuracy. The regularized model uses 4 features, which causes less accuracy with the training dataset but allows accuracy on the test data set to increase. In addition, it takes 0.2133 seconds to graph the baseline model and 0.2004 seconds to graph the regularized model.

In Table 2, both the baseline and regularized model maintain a high accuracy. It follows the trend that the regularized model accuracy, which is 98.74%, on the training set is lower than the baseline model accuracy, which is 99.85%, and how the accuracy on the test datasets are higher for the regularized model rather than the baseline accuracy besides Day 0 dataset with 97.94% for the regularized model’s accuracy and 98.06% for the baseline model’s accuracy. The regularized and baseline accuracies for Day 1, 100, 120 and the average of the 4 days training datasets respectively are 97.53% and 97.47%, 95.97% and 95.40%, 94.71% and 93.54%, and 96.12% and 96.54%. The following days after Day 0 follow the trend in the synthetic dataset. Meanwhile as the Days increase, the accuracy of the logistic regression model, regardless of the baseline model or regularized model, become less accurate.

The percent errors of Day 0 to Day 120, including the average of the test follow the trend seen in the synthetic data where the baseline model’s % error is greater than the regularized

	B Accuracy	R Accuracy	Time / B % Error	Time / R % Error
Day 0 Train	99.85%	98.74%	7.6678s	7.2906s
Day 0 Test	98.06%	97.94%	1.7884%	0.8138%
Day 1 Test	97.47%	97.53%	2.3805%	1.2304%
Day 100 Test	95.40%	95.97%	4.4585%	2.8024
Day 120 Test	93.54%	94.71%	6.3170%	4.0837%
Average of Tests	96.12%	96.54%	3.7361%	2.2326%

Table 2: Baseline (B) and Regularized (R) Accuracy and % Error of Model and time required to train model using Anonymized URL Dataset

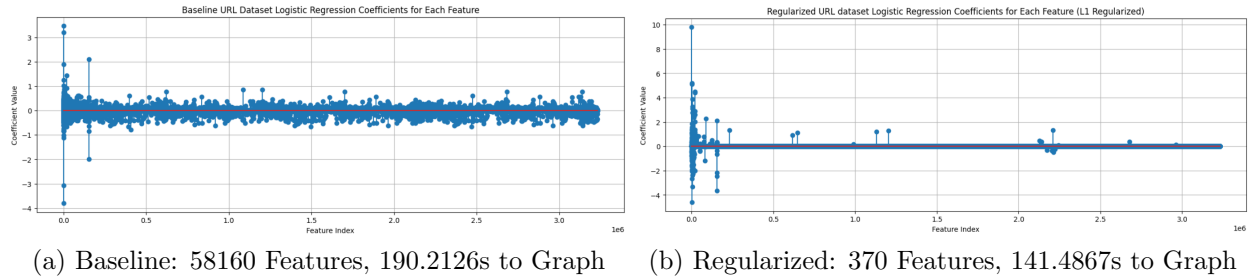


Figure 2: URL Data Coefficient Graphs

model’s % error. The regularized % error and the baseline % error for Day 0, 1, 100, 120, and the average of the test datasets are as follows with 0.8138% and 1.7884%, 1.2304% and 2.3805%, 2.8024%, 4.4585%, 4.0837% and 6.3170%, and 2.2326% and 3.7361%. The baseline % error for the Day 120 test is significantly inaccurate, being over 5%. If the days continued past Day 120, the model would likely worsen. The regularized model would likely follow this but would likely have a few more days before it became inadequate for predicting whether a URL was malignant or benign over 95% of the time.

According to Figure 2, the baseline model uses 58160 features while the regularized model uses 370 features. Included in the 58160 features, many features are noise and causing the model to overfit as seen when comparing to the 370 features, regularized model which have higher accuracy compared to the baseline model’s accuracy as the days increase. In addition, it took 190.2126 seconds to graph the baseline model and 141.4867 seconds to graph the regularized model.

5. Discussion

From the results, three main evaluations can be made on the weaknesses of logistic regression with many features: overfitting, poor generalization, and computational intensity.

The synthetic data clearly show the problem with high features. The trained dataset has high accuracy of 100% while the test data has low accuracy with 66.67%. Meanwhile, when the features are regularized using the lasso to keep only the important features instead of noise, the training accuracy decreases to 88.57% but the test accuracy increases to 76.67%. Rather than having the model have a high accuracy on the training dataset, it is better to have a higher accuracy on the test dataset. However, it is likely the lasso likely caused the model to underfit, so to create a better model more features would likely still be needed to be included.

While the model of the anonymized URL dataset also has multiple features with around 3.2 million, 58160 features were chosen using the same methodology as seen in Table 2 and Figure 2. If all the features were chosen it is likely the model trained on the Day 0 training dataset would reach 100%. Even though the model is trained to only 99.85% accuracy on the training dataset, the model still shows the trends that we saw in the synthetic dataset to less of an exaggerated degree. As the days increased, the accuracy went down as seen in Table 1.

In both synthetic dataset and the real-world anonymized URL dataset, the regularized model was a lower model training accuracy but a higher testing accuracy. In Table 2, the tests on the four different days show a corresponding higher accuracy, besides Day 0, and a higher % error for all. In addition, the regularized data only uses 370 features.

Logistic regression has many limits when met with a dataset with many features. As seen in the more exaggerated results of synthetic dataset and the real world data set that follows the trend, logistic regression overfit if not regularized. This causes poor generalization to the rest of the dataset.

In addition, with more features, there is a problem of computational intensity. Accepting more features seem to be a slight detriment in creating the baseline over the regularized models, with the synthetic data taking 0.2464 seconds and 0.0076 seconds for the baseline and regularized model respectively. With the real-world data, it takes 7.6678 seconds and 7.2906 seconds. In addition, the fitting of the model and creating of the graphs for the baseline of 1000 features took 0.2133 seconds, which is more than the regularized model graph of 0.2004 seconds with 4 features. With the real-set data, the baseline model graph of real-world dataset took 190.2126 seconds to compute, which is more than the regularized model graph taking 141.4867 seconds to compute the graph.

Works Cited

Ma, Justin, et al. "URL Reputation." UCI Machine Learning Repository, 2009, <https://doi.org/10.24432/C5H89Q>. Accessed 19, Apr. 2026.

Acknowledgement of Assistance

Gemini. Assistance given to the author, AI. AI was used to help set up and execute a demonstration to illustrate how logistic regression can overfit when given many features and few observations, and then successfully showed how regularization can mitigate this overfitting, leading to improved model performance on unseen data. It was used to and do data processing for synthetic data and Day 0, 1, and 100. For Day 120, I used the previous code as reference to test both the baseline model and regularized model on Day 120 data. Then, I copied the code over to a new Colab to understand how to code interacts with one another. <https://colab.research.google.com/drive/1ludFq3bkw4sJ8tPqcWN4U2xnfBhCvU8>*scrollTo =_s5Wge2n2NNa.WestPoint, NY, 29APR*